

#05 AWS Textract Document Extraction Agent

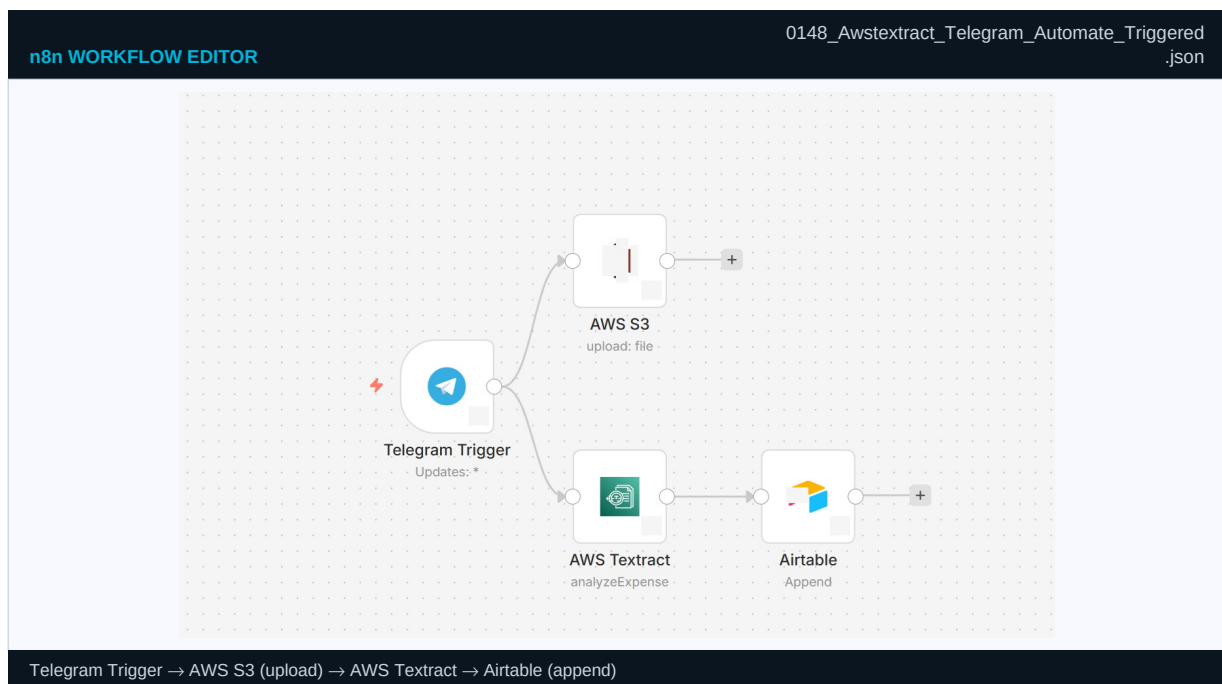
AWS / Intelligent
Document
Processing

Photo an invoice on Telegram — structured data in Airtable in 30 seconds

BUSINESS PROBLEM

Finance teams drown in paper invoices requiring manual data entry. This workflow lets staff photo any invoice with their phone, send to a Telegram bot, and AWS Textract extracts all key-value pairs — vendor, amount, date, PO number — into structured Airtable records automatically.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Telegram message — photo sent by finance staff to bot

```
{
  "message_id": 4821,
  "from": {
    "id": 112233,
    "username": "ahmed.finance"
  },
  "photo": [
    { "file_id": "AgACAgQ...xyz", "width": 1280, "height": 960 }
  ],
  "caption": "Supplier invoice June batch",
  "date": 1718425200
}
```

OUTPUT — Airtable record created — structured invoice fields

```
{
  "json": {
    "vendor": "Gulf Office Supplies LLC",
    "invoice_no": "INV-2024-00871",
    "invoice_date": "2024-06-10",
    "due_date": "2024-07-10",
    "amount": "OMR 1,247.50",
    "currency": "OMR",
    "po_reference": "PO-2024-FIN-0441",
    "line_items": 8,
    "confidence": 0.97,
    "s3_url": "s3://textract-intake/2024-06/INV-00871.jpg",
    "status": "Pending Approval"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Telegram Trigger — photo message	Staff sends invoice photo to Telegram bot. Trigger fires with image binary, sender ID, caption, and timestamp.
2	■ AWS S3 — upload for Textract	Image uploaded to S3 bucket: textract-intake/{date}/{uuid}.jpg. Returns S3 URI for the Textract API call.
3	■ AWS Textract — AnalyzeDocument	API call with FORMS + TABLES features. Returns Blocks[] with KEY_VALUE_SET pairs, TABLE cells, and confidence scores.
4	■ Airtable — create structured record	Parsed fields written: vendor, invoice_no, amount, currency, due_date, line_items count, confidence, S3 URL, status.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
AWS Access Key + Secret	AWS Console → IAM → Needs: textract:*, s3:PutObject on intake bucket	AWS Credentials
Telegram Bot Token	Message @BotFather → /newbot → copy token	Telegram API
Airtable Personal Access Token	airtable.com/create/tokens → scope: data.records:write	Airtable API

SAMPLE RUN — what this workflow processed and produced

Scenario	A trading company receives 40-60 supplier invoices per week across 3 offices. Finance staff previously spent 4-5 hours manually entering invoice data into their ERP system.
Input received	Ahmed from the finance team photographed a Gulf Office Supplies invoice and sent it to the company Telegram bot at 09:14. Invoice was a standard A4 document with 8 line items.
What the workflow found	Textract returned 34 key-value pairs. All critical fields extracted with confidence above 0.95. Line items table parsed correctly including unit prices and VAT amounts.
Output produced	Airtable record created in 28 seconds. Record linked to PO-2024-FIN-0441 in the existing procurement table. Finance manager notified via email for approval. Zero manual data entry required.

JSON: 0148_Awstextract_Telegram_Automate_Triggered.json	GitHub: trm-textract-invoice-agent	aws.amazon.com/textract/
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